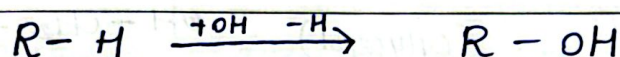


Chapter ~ 7

Alcohols, Phenols & Ethers.

⇒ Alcohols — Classification and Nomenclature.



Classification by degree of carbon bearing.

Type	Example	Structure
1° (Primary)	Ethanol	CH_3-CH_2-OH
2° (Secondary)	Isopropanol (propan-2-ol)	$CH_3-CH(OH)-CH_3$
3° (tertiary)	tert-Butanol	$CH_3-\overset{\overset{OH}{ }}{C}-CH_3$ $\quad \quad $ $\quad \quad CH_3$

General Formula: Saturated acyclic monohydric alcohols
 $C_n H_{2n+1} OH$ or $(C_n H_{2n+2} O)$

⇒ Difference between 1°, 2° and 3° Alcohols.

Property	1° Alcohol	2° Alcohol	3° Alcohol.
i) Structure	-OH attached to only one other C	-OH ^{carbon} attached to two other C	-OH carbon attached to three other C.
ii) Boiling point	Highest	Intermediate	Lowest.
iii) Oxidation	→ Carboxylic acid	→ Ketone	→ No reaction.
iv) Dehydration ease	Slowest	faster	fastest.
v) Lucas reagent	No turbidity	Turbid in mins	Turbid instantly
vi) SN ₂ substitution	Fastest	Slower	Slowest

⇒ Classification by number of -OH groups.

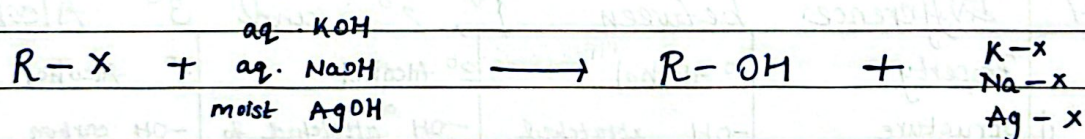
- i) Monohydric Alcohol — one -OH groups
(eg Ethanol)
- ii) Dihydric Alcohol — two -OH gps : $\text{OH} - \text{CH}_2 - \text{CH}_2 - \text{OH}$
(ethylene glycol)
- iii) Polyhydric Alcohol — three -OH groups :
(Glycerol) $\text{OH} - \text{CH}_2 - \overset{\text{OH}}{\underset{|}{\text{CH}}} - \text{CH}_2\text{OH}$

⇒ Special Structure Classes.

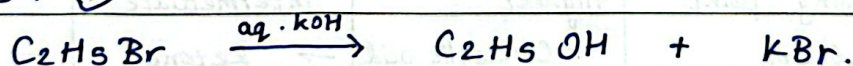
- Allylic Alcohol : -OH on carbon adjacent to a C=C double bond.
 $\text{CH}_2 = \text{CH} - \overset{\text{OH}}{\text{CH}_2} - \text{OH}$. (prop-2-en-1-ol).
- Benzylic Alcohol :  (Benzyl Alcohol)

⇒ Methods of Preparation of Alcohols.

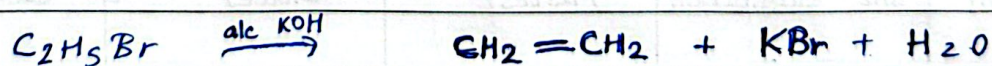
(a) From Alkyl Halides. — Hydrolysis



Example : ①

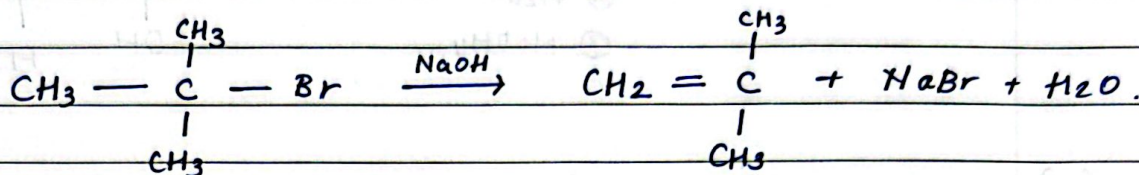


Example : ②



Relative Tendency

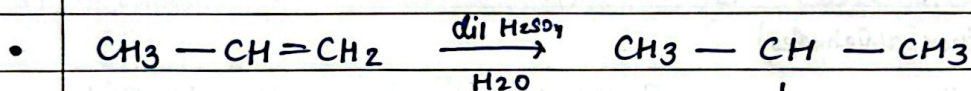
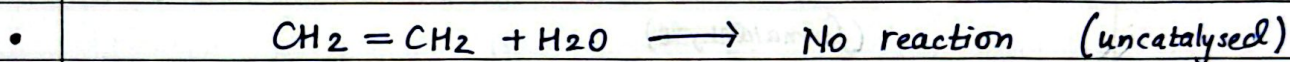
- S_N2 Substitution : $1^\circ \text{ Alc} > 2^\circ \text{ Alc} > 3^\circ \text{ Alc}$
- Elimination (Alkene) : $3^\circ \text{ Alc} > 2^\circ \text{ Alc} > 1^\circ \text{ Alc}$



(b) From Alkenes. — (Hydration).

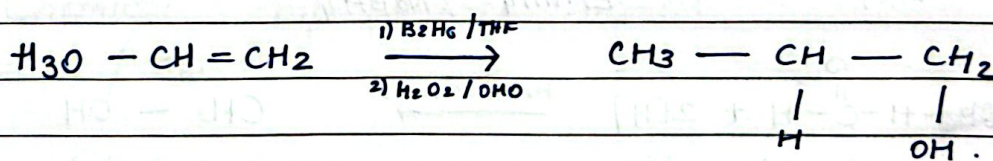
i) Acid catalysed direct hydration

Addition of $\text{H}-\text{OH}$ needs an acidic catalyst.

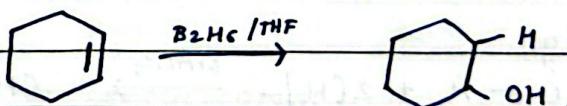


Markownikoff's Rule.

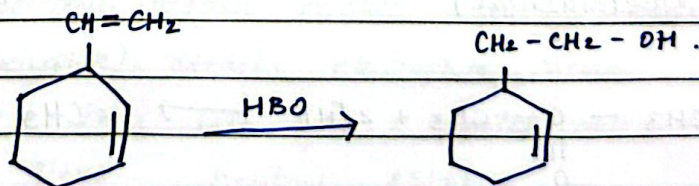
ii) Hydroboration — oxidation (anti-markownikoff, indirect hydration).



Example : ①

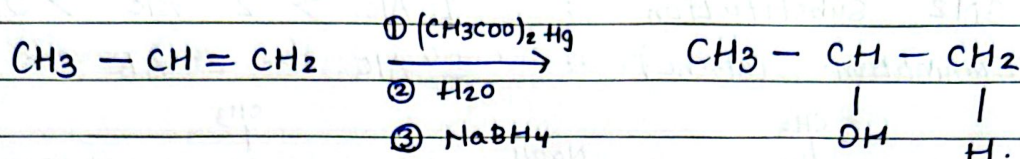


Example : ②

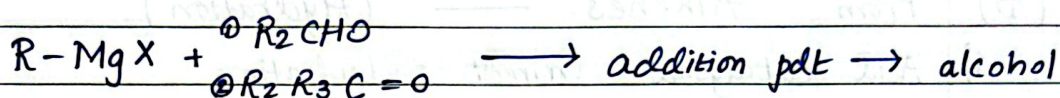


- Anti-markownikoff Rule
- Rate $\propto 1 / \text{Steric hindrance}$
- BH_3 is unstable and becomes B_2H_6

iii) Oxymercuration - Demercuration (Markovnikov, addition of H_2O)



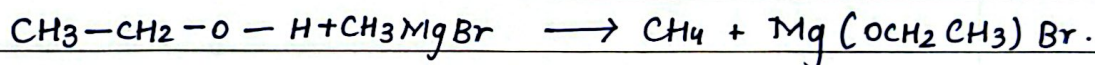
(C) From Grignards Reagent.



Example : ① $HCHO + R-MgX \longrightarrow 1^\circ \text{ Alcohol}$.
(formaldehyde)

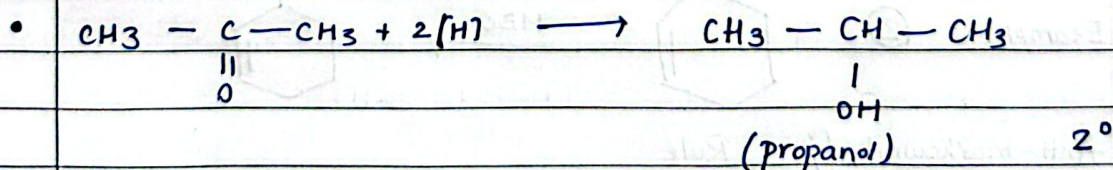
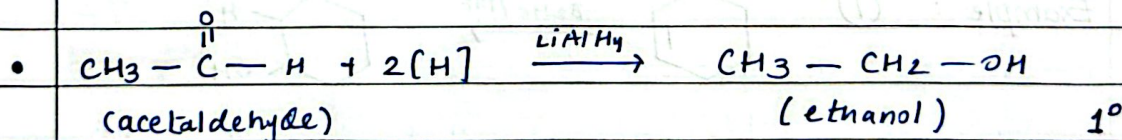
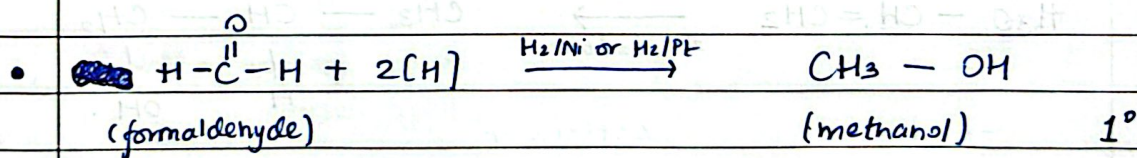
② $R_2CHO + R-MgX \longrightarrow 2^\circ \text{ Alcohol}$
(other aldehydes)

③ $R_2R_3C=O + R-MgX \longrightarrow 3^\circ \text{ Alcohol}$



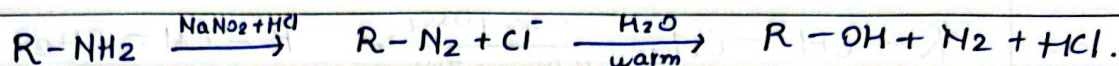
(d) Reduction of Carbonyl compounds.

Reagents : H_2/Ni , H_2/Pt , H_2/Pd or
 $LiAlH_4$ / $NaBH_4$



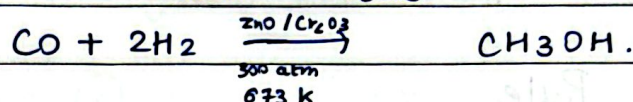
Note: 3° Alcohols can not be prepared by this

(e) From Primary Amines.

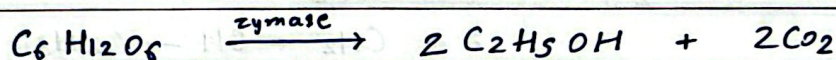
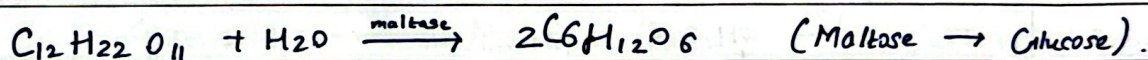
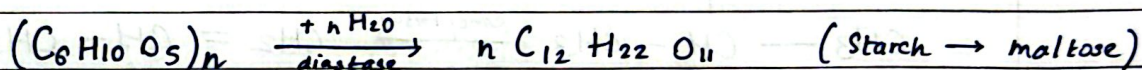


(f) Manufacture of Methanol and Ethanol.

i) Methanol - Bosch Process

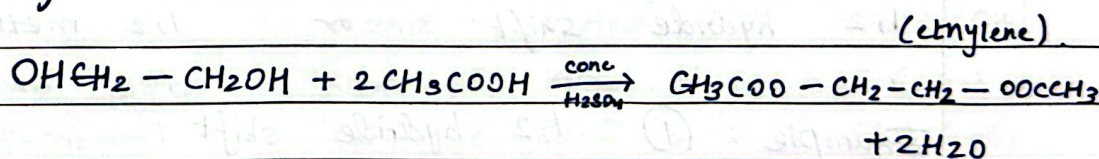


ii) Ethanol - Fermentation of carbohydrates.



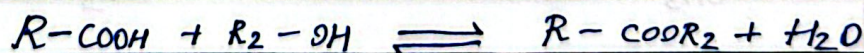
⇒ Chemical Properties of Alcohols.

(a) Esterification

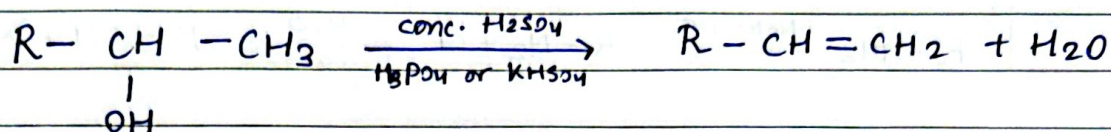


Mechanism: (Fischer esterification)

1. Protonation of carbonyl oxygen makes carbonyl carbon electrophile
2. The alcohol (nucleophile) attacks electrophile carbon
3. Proton transfer, then loss of a water molecule → carbonyl.
4. Deprotonation gives neutral ester.



(b) Dehydration of Alcohols

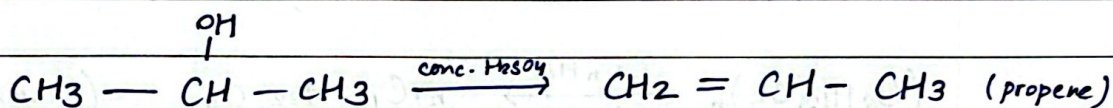


Mechanism

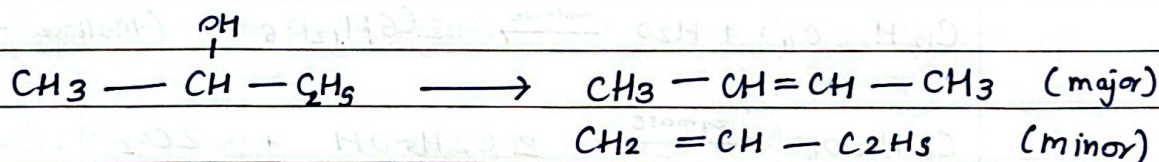
1. (Protonation) $\text{R} - \underset{\text{OH}}{\text{CH}} - \text{CH}_3 \xrightarrow{+ \text{H}^+} \text{R} - \underset{\text{OH}_2^+}{\text{CH}} - \text{CH}_3$
2. (loss of water) $\text{R} - \underset{\text{OH}_2^+}{\text{CH}} - \text{CH}_3 \xrightarrow{- \text{H}_2\text{O}} \text{R} - \text{CH} - \text{CH}_3$
3. (loss of B-H alkene) $\text{R} - \text{CH} - \text{CH}_3 \longrightarrow \text{R} - \text{CH} = \text{CH}_2$

=> Saytzeff's Rule

Example : (1)



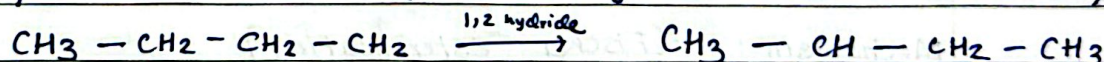
(2)



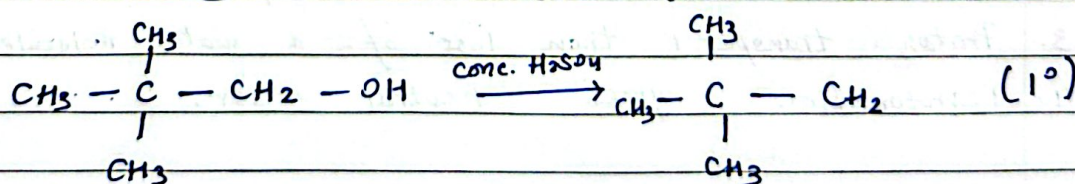
=> Carbocation rearrangement during dehydration.

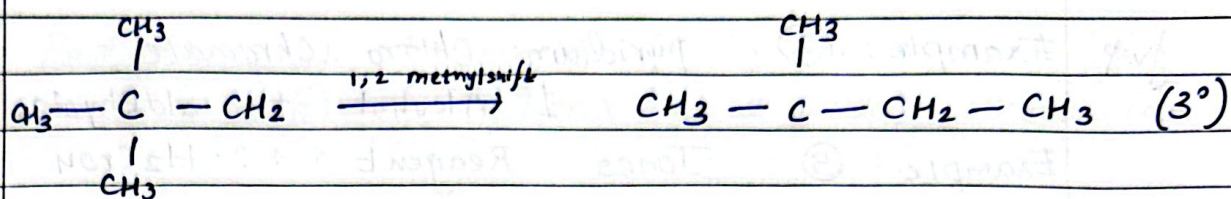
When it is not stable it shifts to
1,2 hydride shift or 1,2 methyl shift

Example : (1) 1,2 hydride shift (1° - 2°)



Example : (2) 1,2 methyl shift

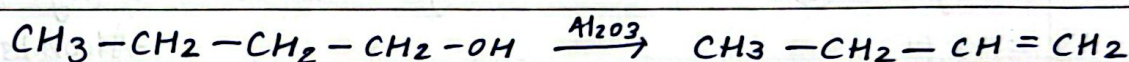




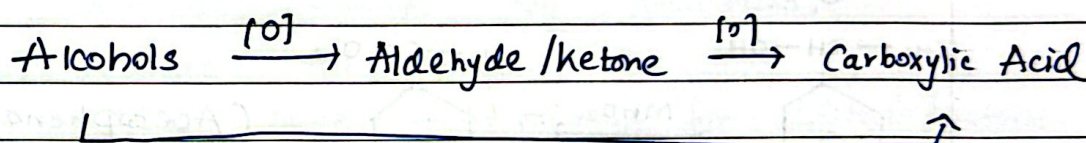
Important Exception — where no carbocation occurs

" If the dehydrating agent is Al_2O_3 , P_2O_5 or POCl_3 instead of conc. H_2SO_4 no carbocation or rearrangement occurs, only Saytzeff rule

Example: (1)



(c) Oxidation of Alcohols



Product depends on

- i) Type of Alcohol
- ii) type of reagent.

Example: (1) with Acidic $\text{K}_2\text{Cr}_2\text{O}_7$ or KMnO_4 and Red hot tube (Cu)

Alcohol type	KMnO_4 / Acidic $\text{K}_2\text{Cr}_2\text{O}_7$	Red hot tube
1° ($\text{R}-\text{CH}_2-\text{OH}$)	carboxylic Acid $\text{R}-\text{COOH}$	Aldehyde $\text{R}-\text{CHO}$
2° ($\text{R}-\underset{\text{OH}}{\text{CH}}-\text{R}$)	Ketone $\text{R}-\text{CO}-\text{R}$	Ketone $\text{R}-\text{CO}-\text{R}$
3° Alcohol	No reaction	Alkene (loss of H_2O).

Example: (2) Pyridium Chloro Chromate
1° Alcohol to aldehydes only

Example: (3) Jones Reagent : H_2CrO_4

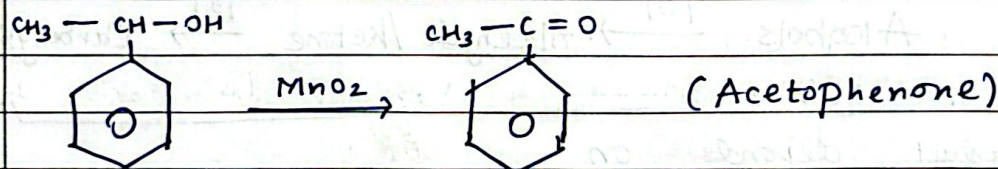
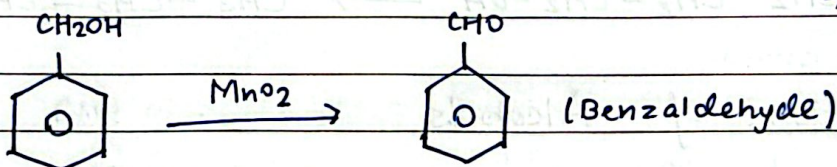
Example: (4) Collins Reagent : CrO_3 : pyridine : CH_2Cl_2
1° Alcohol to aldehydes

Example: (5) Sarrets Reagent :
1° Alcohol to aldehydes

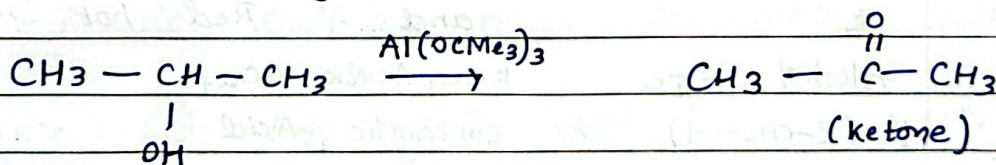
Example: (6) NBS : allylic / Benzylic bromination.

Example: (7) Haloform

Example: (8) MnO_2 : only oxidises Benzylic / Allylic Alcohols.

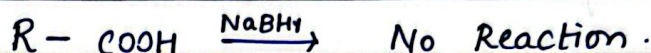
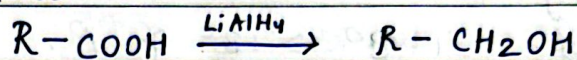


Example: (9) OPR Oxidation (Oppenauer's oxidation)
Only works on 2° Alcohols

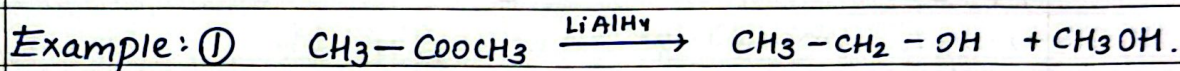
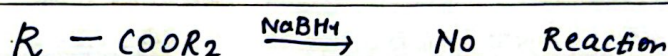
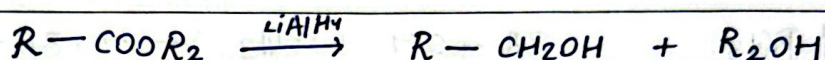


(d) Reduction of Carboxylic Acids
 RCOOH (acid) / RCOX (acid halide) / RCOOR_2 (ester)

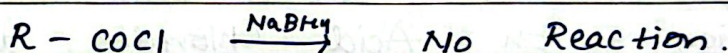
1) Carboxylic Acid :



2) Ester :



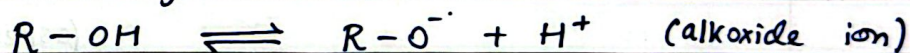
3) Acid Chloride :



LiAlH_4 reduces All three.

NaBH_4 reduces none $\rightarrow$ reserved for aldehyde/ketone.

(e) Acidic Nature of Alcohols:

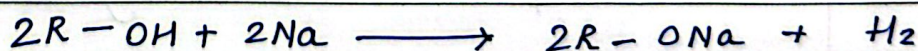


Alcohols are less acidic than water.

and much less acidic than phenols.

Acidity order : $1^\circ_{\text{alc}} > 2^\circ_{\text{alc}} > 3^\circ_{\text{alc}}$.

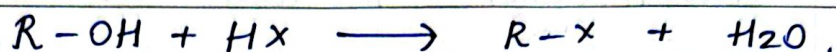
(f) Reaction with Sodium.



(sodium alkoxide)

Brisk effervescence.

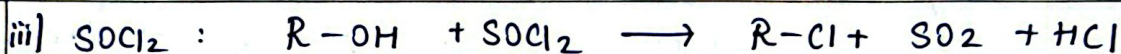
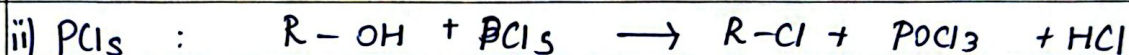
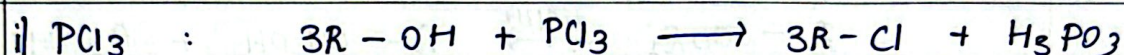
(g) Reaction with hydrogen halides (HX)



Reactivity of : $HX: HI > HBr > HCl.$

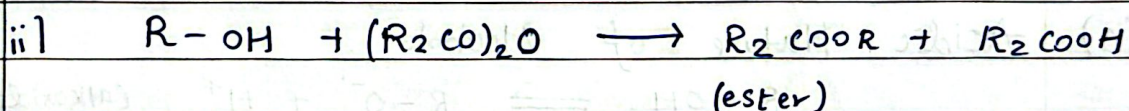
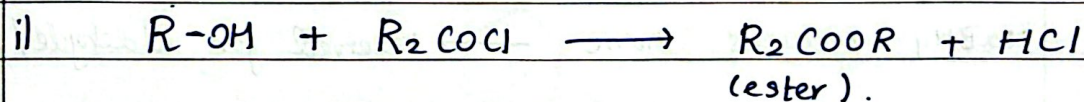
Reactivity of OH : $3^\circ > 2^\circ > 1^\circ.$

(h) Reaction with PCl_3 , PCl_5 and $SOCl_2$.



(Darzens process).

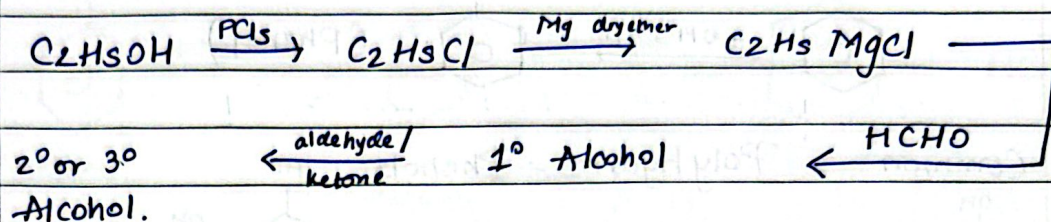
(i) Reaction with Acid Chlorides and Acid Anhydrides



$\Rightarrow$ Uses of Alcohols.

- Methanol : solvent, denaturant for ethanol
possible alternative fuel, additive.
- Ethanol : solvent, in beverages, as an antiseptic, as a fuel
- Generally, preparation of dyes, drugs, perfumes, preservative.

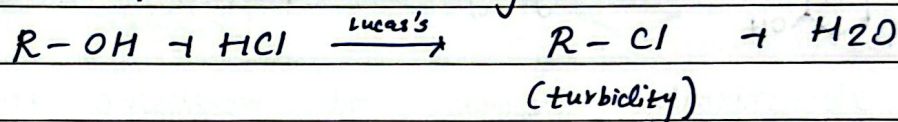
⇒ Conversion of one alcohol into another.



⇒ Distinction between 1° and 2° and 3° Alcohols
Lucas's Test.

Lucas's reagent : anhydrous ZnCl_2 + conc. HCl .

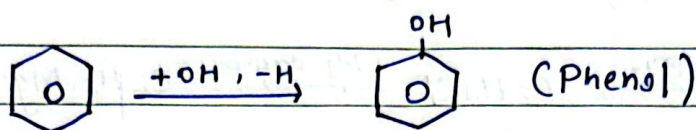
Principle : Alcohol $\longrightarrow$ Alkyl Chloride



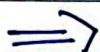
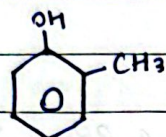
Time of turbidity : $1^\circ > 2^\circ > 3^\circ$ fastest.
no turbidity

⇒

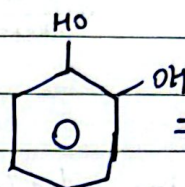
⇒ Phenols (Classification, Nomenclature).



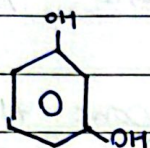
• Common Polyhydric Phenols



o-Cresol



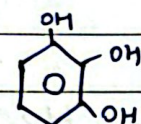
Catechol



Resorcinol



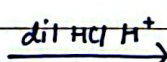
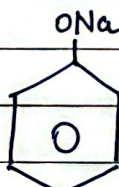
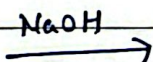
Quinol



Pyrogallol.

⇒ Preparation of Phenols.

(a) From Haloarenes — Dow's Process

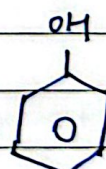
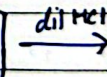
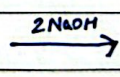
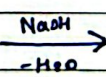
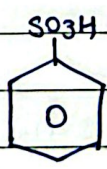
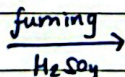


Chlorobenzene

sodium phenoxide

Phenol.

(b) From Benzene Sulphonic Acid.

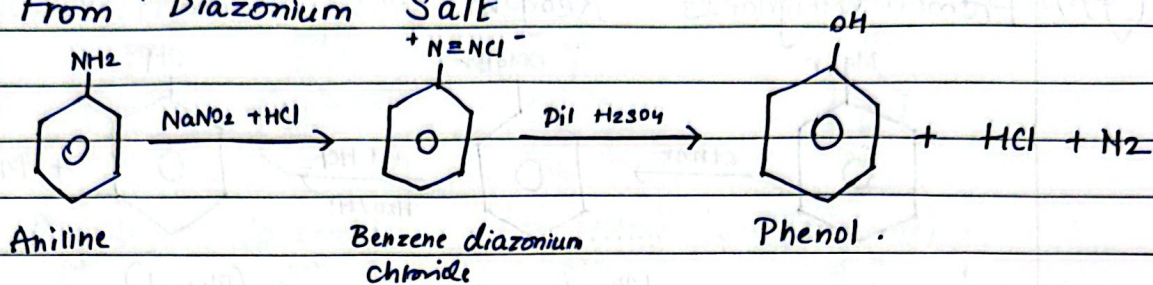


Benzene

Benzene sulphonic acid

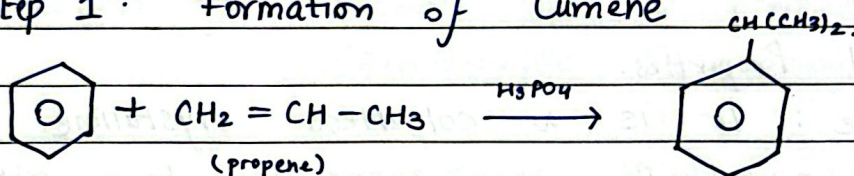
(Phenol)

(c) From Diazonium Salt

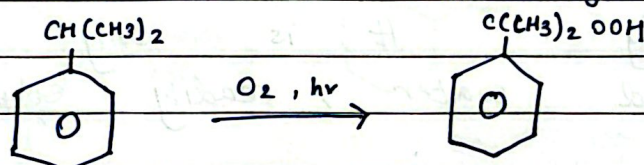


(d) From Cumene — Industrial Process.

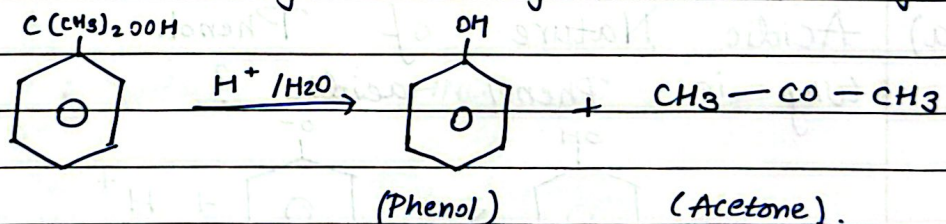
Step 1: Formation of Cumene



Step 2: Oxidation to cumene hydroperoxide

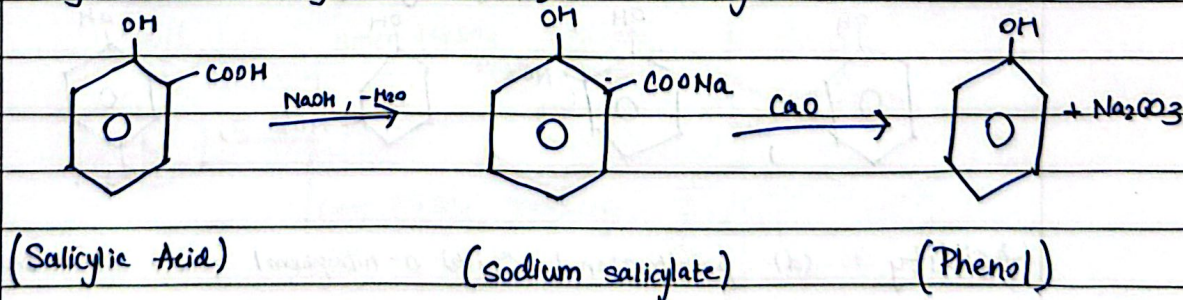


Step 3: Acid Catalysed Cleavage (Hock's rearrangement).

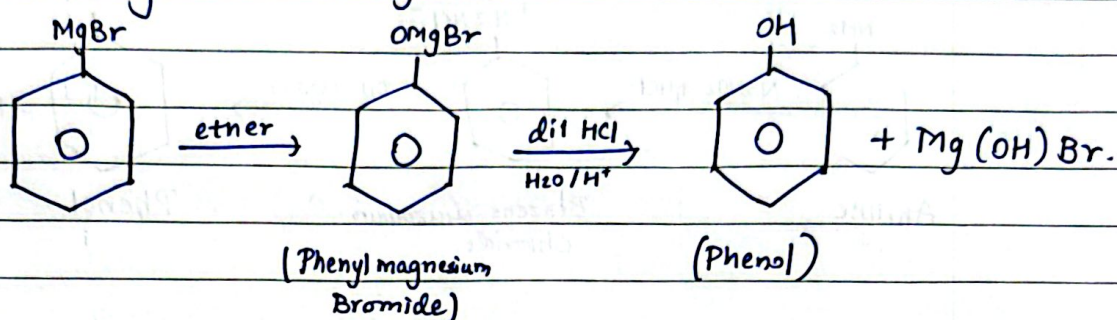


This is a major industrial process of Phenol.

(e) By decarboxylation of Salicylic Acid.



(f) From Grignard's Reagent.



⇒ Physical and Chemical Properties of Phenols.

• Physical Properties.

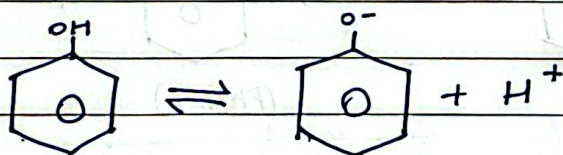
a) State : It is a colourless crystalline solid. turns pink/red on exposure to air with a sharp medicinal odour.

b) Solubility : It is only slightly soluble in cold water, readily soluble in organic solvents.

• Chemical Properties of Phenol.

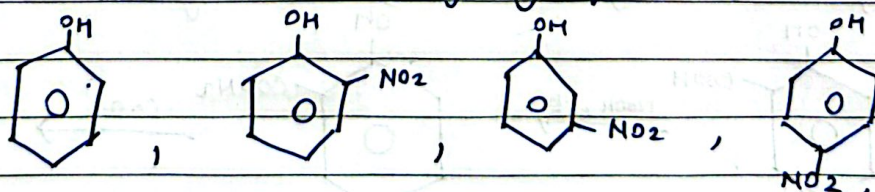
(a) Acidic Nature of Phenol.

Why is Phenol acidic?



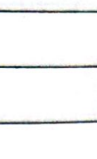
Phenoxide ion

ii) Effect of electron withdrawing groups (increase acidity).



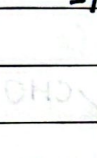
Acidity : (d) p-nitrophenol > (b) o-nitrophenol > (c) m-nitrophenol.

10

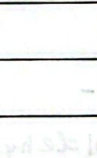


(b)

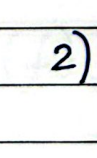
1)



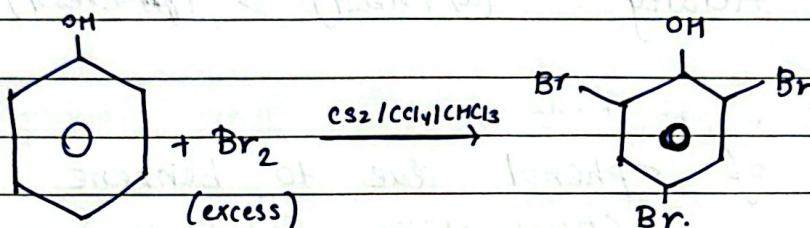
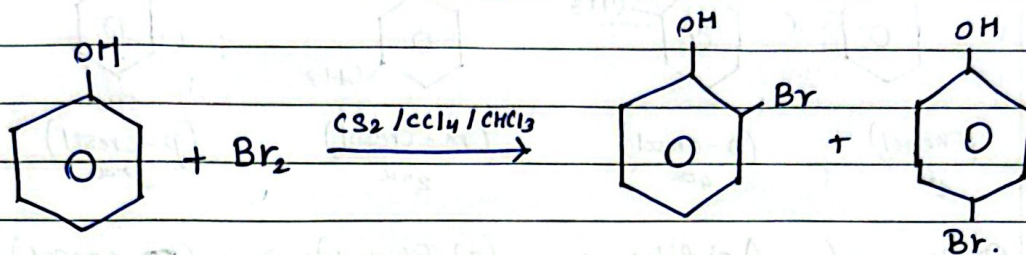
10



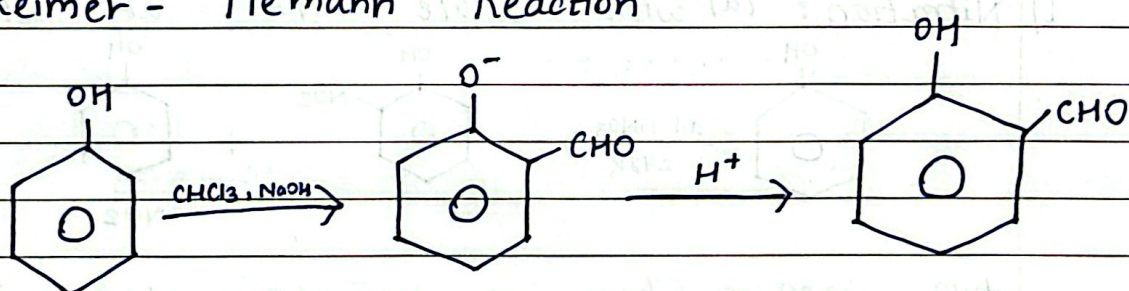
2)



3) Halogenation : with a less polar solvent ($\text{CS}_2 / \text{CCl}_4 / \text{CHCl}_3$).



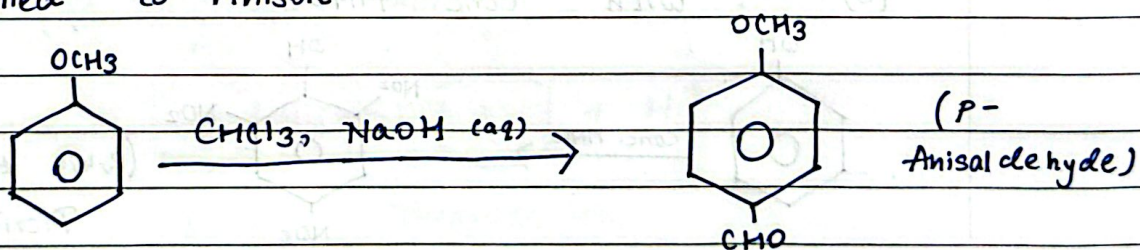
(c) Reimer - Tiemann Reaction
Imp.



(Phenol)

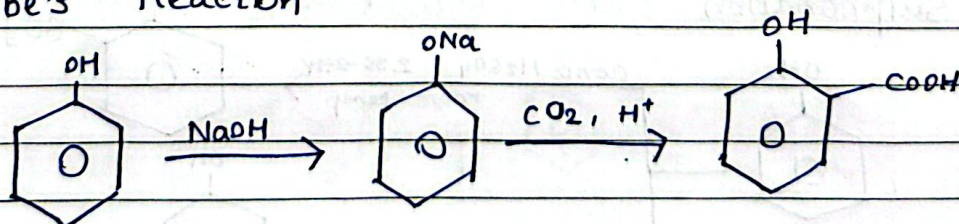
(Salicylaldehyde)

• Applied to Anisole



(p-Anisaldehyde)

(d) Kolbe's Reaction

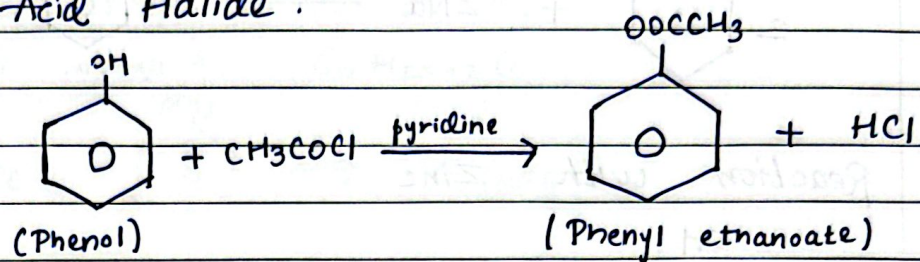


(Phenol)

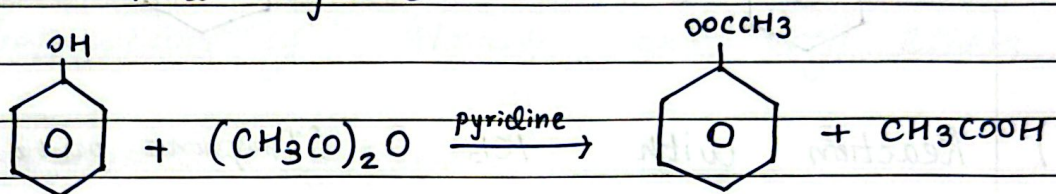
(Salicylic acid)

(e) Esterification of Phenol.

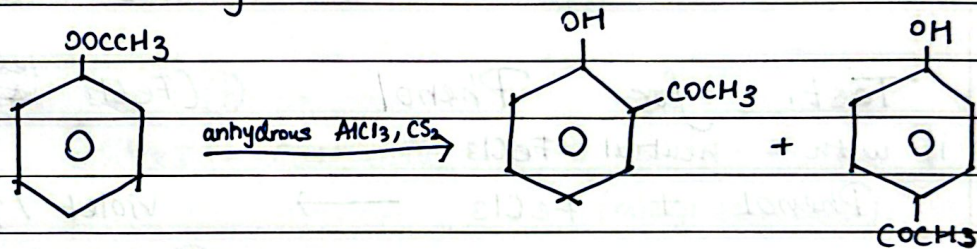
1) with Acid Halide:



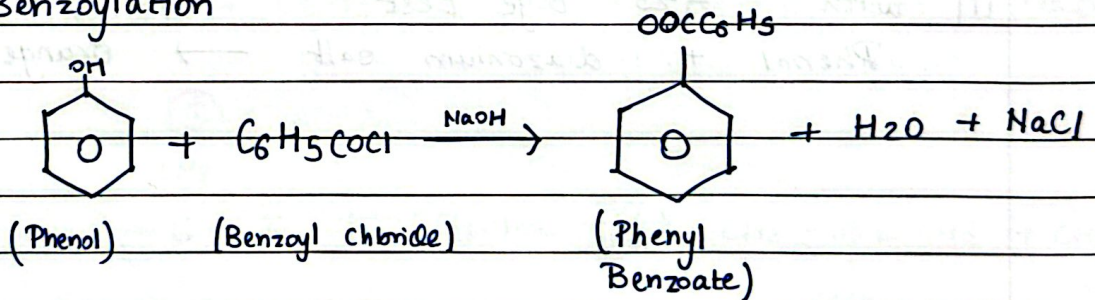
2) with Acid anhydride



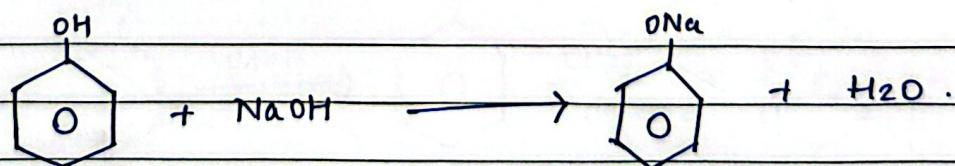
3) Fries Rearrangement



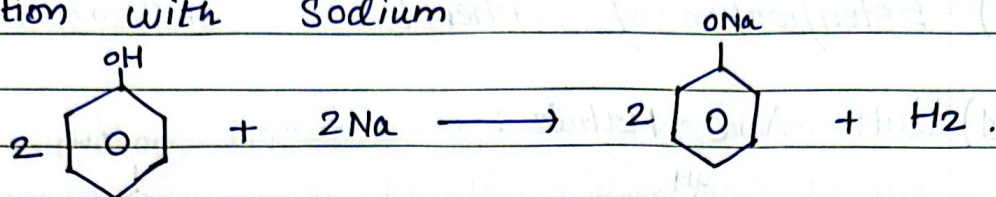
4) Benzoylation



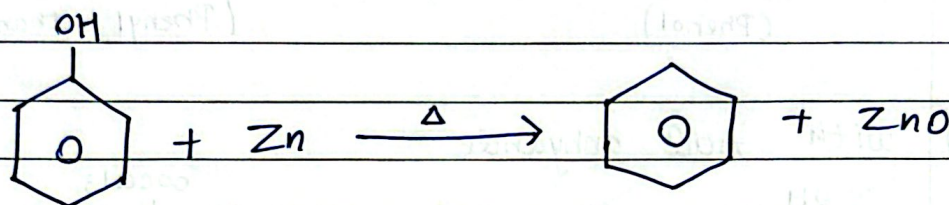
(f) Reaction with Sodium hydroxide



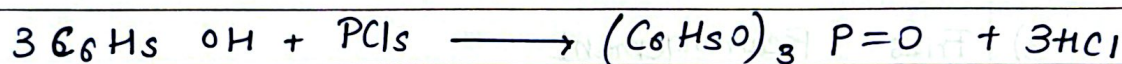
(g) Reaction with Sodium



(h) Reaction with Zinc

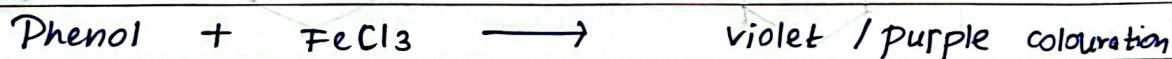


(i) Reaction with PCl_5 (Phosphorus pentachloride)

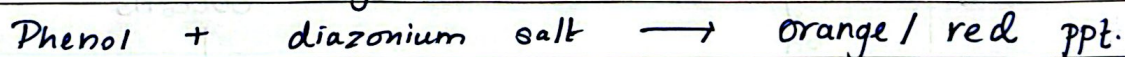


(j) Test for Phenol (FeCl_3 test)

i) with neutral FeCl_3



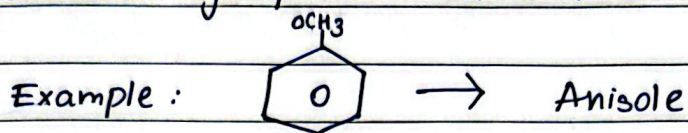
ii) with Azo dye test



⇒ Ethers — Structure & Nomenclature

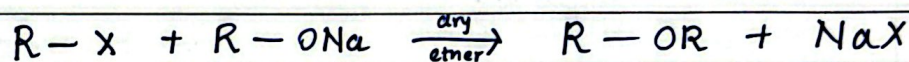
Ethereal group : $R-O-R_2$

General group : $C_n H_{2n+2} O$

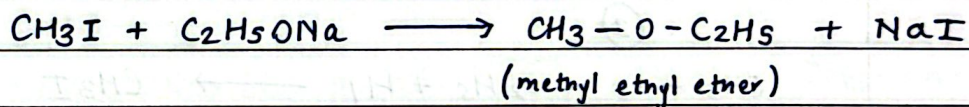


⇒ Preparations of Aliphatic and Aryl ethers

(a) Williamson's Synthesis

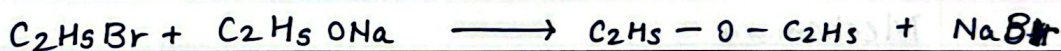


→ Examples : ①



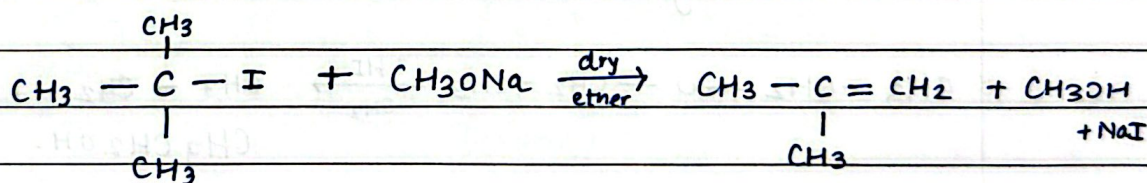
→

②



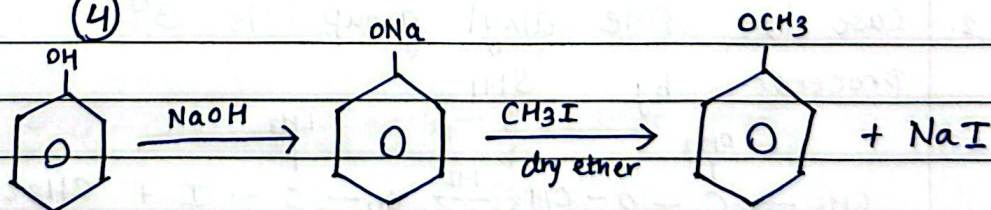
→

③

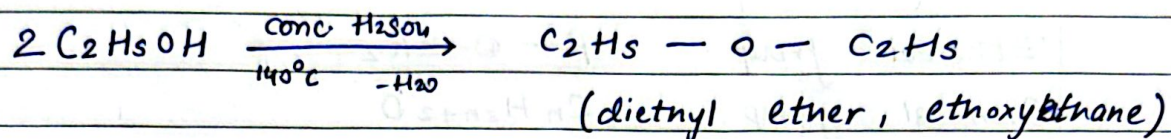


→

④

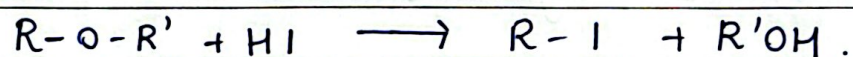


(b) Dehydration of Alcohols



⇒ Properties of Ethers

(a) Reaction with HI (Cleavage of C-O bond) Aliphatic Ether



Example : ①

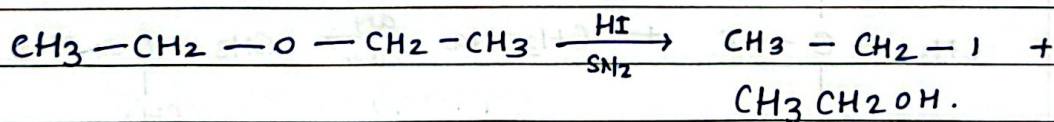


②

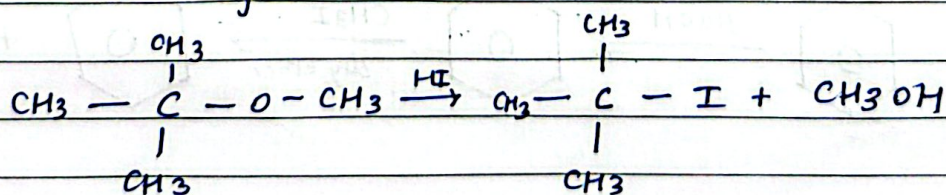


Mechanism

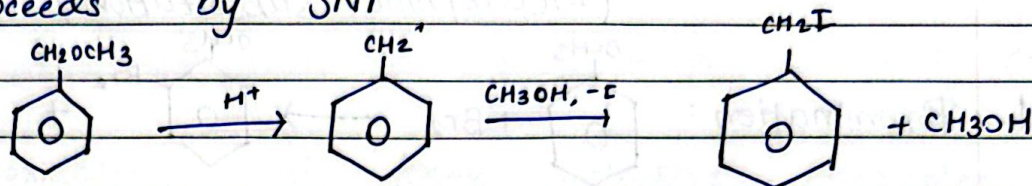
- Case (i) : both alkyl groups are 1° : Reactions proceed by $\text{S}_\text{N}2$



- Case (ii) : one alkyl group is 3° : Reaction proceeds by $\text{S}_\text{N}1$



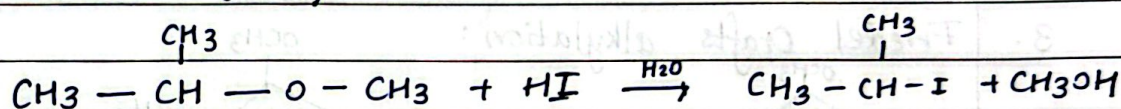
3. Case (iii) : One Alkyl Group is Benzyl : Reaction proceeds by S_N1



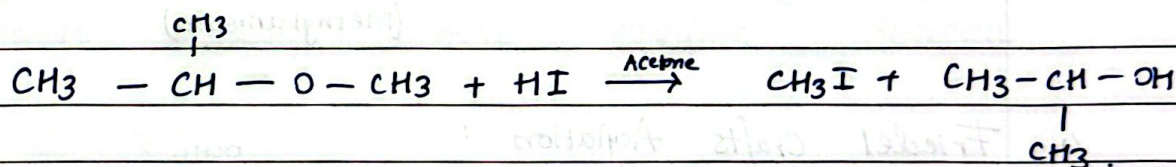
4. Case (iv) : One alkyl group is 2° : Then the outcome depends on solvent

- in protic solvent (H_2O or $\text{C}_2\text{H}_5\text{OH}$) by S_N1
- in acetone by S_N2 .

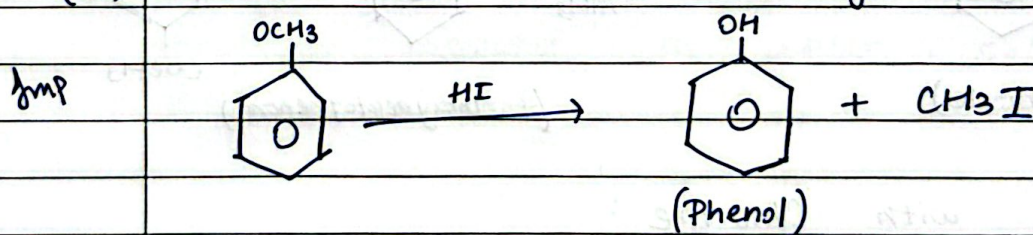
(a) By H_2O (S_N1)



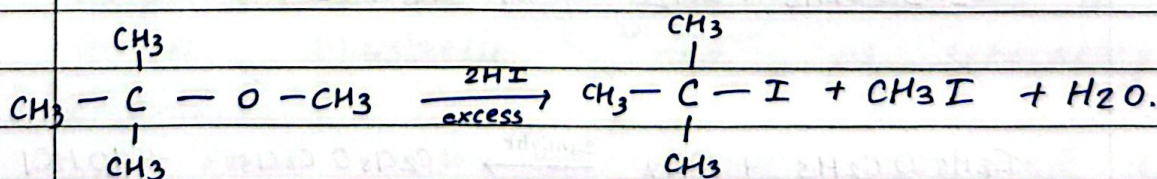
(b) By Acetone (S_N2)



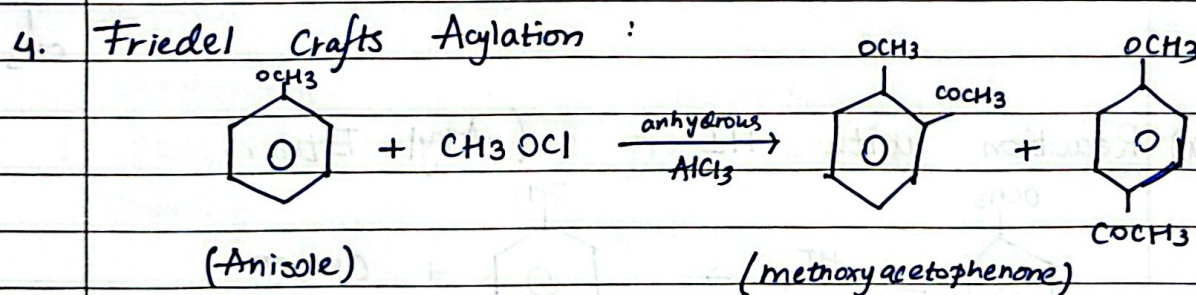
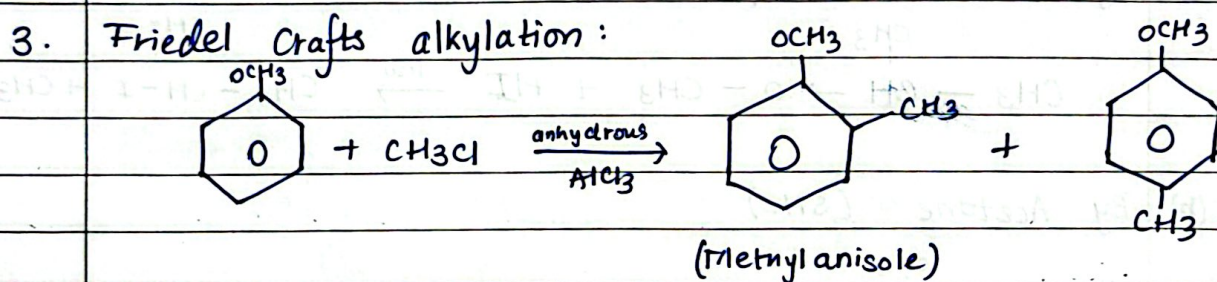
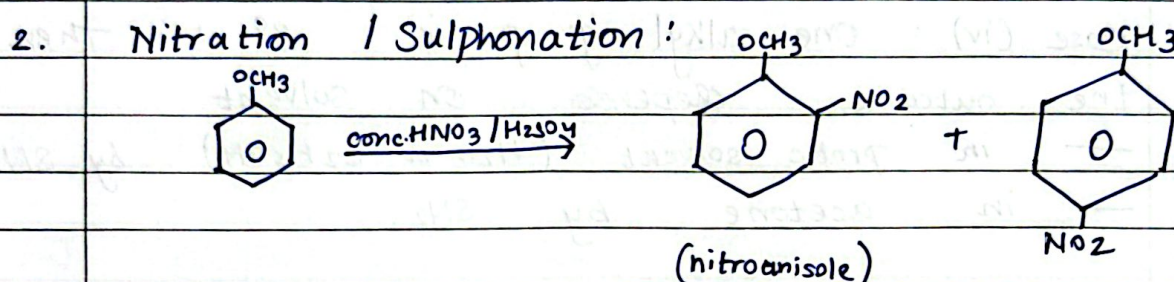
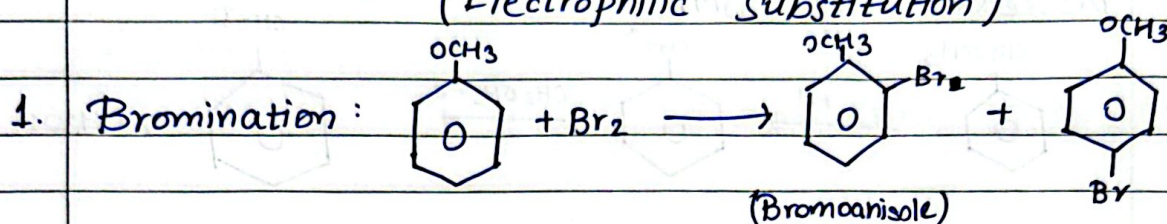
(a) Reaction with HI (Aryl Ether).



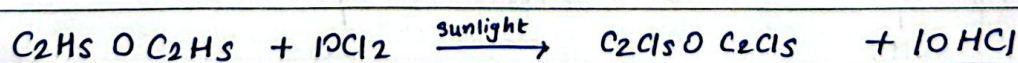
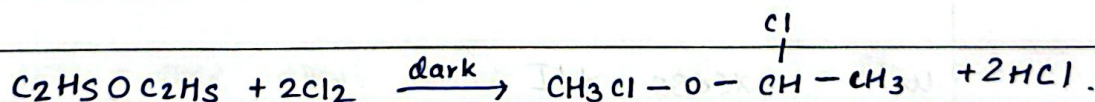
With excess HI



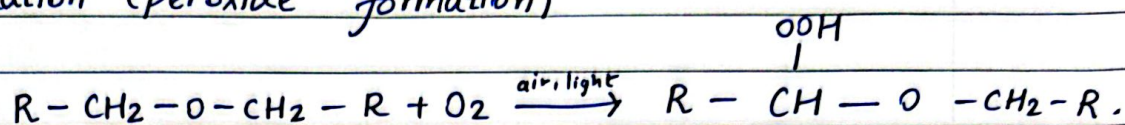
(b) Reactions of Aryl Ethers due to Benzene ring (Electrophilic Substitution)



(c) Reaction with Chlorine:

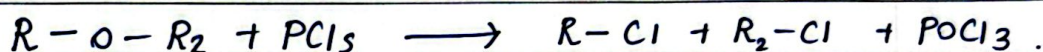


(d) Oxidation (peroxide formation)



Autoxidation $\Rightarrow$ forms explosive peroxides.

(e) Reaction with PCl_5 .



$\Rightarrow$ Physical Properties of Aliphatic and Aryl Ether.

i) Aliphatic ethers:

State: lower ethers (dimethyl ether) are gases
common ethers are volatile, only slightly soluble.

Miscible: with most organic solvents.

ii) Aryl Ethers.

state: Anisole and similar ethers are colourless liquids with pleasant odour.

practically insoluble in water, soluble in organic solvents.

$\Rightarrow$ Uses of Ethers.

Diethyl ether was generally used as a general anaesthetic and as laboratory solvents.

Aryl ethers such as anisole are used as intermediates in perfumes.